

Name
Class
R. No.

• Mathematics • IIT JEE
• Test 5(July 22 , 2010) • Functions : Bijectivity, Composition
• Maximum Marks 70 • Time 50 minutes

Instructions

1. The test contains two sections containing 8 questions.
2. The marking scheme will include marks for partially correct answers also.

Section I

Q1: Show that the function $f: N \rightarrow N$, given by $f(x) = 2x$, is one-one but not onto

4 Marks

Q2: Prove that the function $f: R \rightarrow R$, given by $f(x) = 2x$, is one-one and onto.

4 Marks

Q3: Show that $f: N \rightarrow N$, given by

$$f(x) = \begin{cases} x + 1, & \text{if } x \text{ is odd} \\ x - 1, & \text{if } x \text{ is even} \end{cases}$$

Is both one-one and onto.

8Marks

Q4: Let A and B be sets. Show that $f: A \times B \rightarrow B \times A$ such that $f(a, b) = (b, a)$ is a bijective function.

8 Marks

Section II

Q5: Show that if $f: A \rightarrow B$ and $g: B \rightarrow C$ are one-one, then $gof: A \rightarrow C$ is also one-one.

8 Marks

Q6: Show that if $f: A \rightarrow B$ and $g: B \rightarrow C$ are onto, then $gof: A \rightarrow C$ is also onto .

8 Marks

Q7: Prove that gof is one-one implies that f is one-one and gof is onto implies that g is onto. 20 Marks

Q8: Prove that if $f: X \rightarrow Y$ is a function such that there exists a function $g: Y \rightarrow X$ such that $gof = I_X$ and $fog = I_Y$, then f must be one-one and onto.

10 Marks